Cumulant Generating Function

1 Cumulant Generating Function (CGF)

For some theoretical distributions it is more convenient to get a suitable form of $\ln M_X(t)$ instead of $M_X(t)$. We write,

$$\ln M_X(t) = \sum_{r=1}^{\infty} k_r \frac{t^r}{r!}$$

Here, $\ln M_X(t)$ is known as CGF and is denoted by $K_X(t)$. The coefficient of $\frac{t^r}{r!}$ in the expansion of $K_X(t)$ is the rth order cumulant.

2 Relationship with MGF

We have,

$$M_{X-\mu}(t) = e^{-\mu t} M_X(t)$$
$$K_X(t) = \ln M_X(t)$$
$$M_X(t) = e^{K_X(t)}$$
$$\Rightarrow M_X(t) = e^{\mu t}\left(1 + \frac{t^2}{2!}\mu_2 + \frac{t^3}{3!}\mu_3 + \cdots\cdots\right)$$
$$\Rightarrow K_X(t) = \mu lnn\left(1 + \frac{t^2}{2!}\mu_2 + \frac{t^3}{3!}\mu_3 + \cdots\right)$$
$$\Rightarrow K_X(t) = \mu t + \left(\tfrac{t^2}{2!}\mu_2 + \tfrac{t^3}{3!}\mu_3 + \cdots\right) - \tfrac{1}{2!}\left(\tfrac{t^2}{2!}\mu_2 + \tfrac{t^3}{3!}\mu_3 + \cdots\right)^2 + \cdots$$
$$\Rightarrow M_X(t) = e^{t\mu} M_{X-\mu}(t)$$

$\therefore k_1 =$ coeff of $t = \mu_1 = \mu_1'$
$\therefore k_2 =$ coeff of $\frac{t^2}{2!} = \mu_2$
$\therefore k_3 =$ coff of $\frac{t^3}{3!} = \mu_3$
$\therefore k_4 =$ coeff of $\frac{t^4}{4!} = \mu_4 - 3\mu_1^2$
ie. $\mu_4 = k_4 + 3k_2^2$
Example: (1) $M_X(t) = (1-\theta t)^{-1}$, $|t| < \frac{1}{\theta}$, find $K_X(t)$ and first four moments.
(2) $X \sim \text{Poi}(\lambda)$, find $K_X(t)$ and first four moments.

3 Characteristic Function (CF)

let X be a random variable defined on the probability space $(\Omega, \mathcal{F}, P)$. CF of X is denoted by $\phi_X(t)$ which would be defined as,

$$\phi_X(t) = E\left(e^{itX}\right) = \begin{cases} \int_\Omega e^{itx} dp, & [\text{ lebesgue integral}] \\ \int_R e^{itx} dp_x, & [\text{ LS integral}] \\ \int^{itx} dF_X(x), & [\text{ RS integral }] \end{cases}$$

where $i^2 = -1$

Notice that we can write, $e^{itx} = \cos(tx) + i\sin(tx)$.

$$\Rightarrow \phi_X(t) = E(\cos(tx)) + iE(\sin(tx))$$

4 Observations

1. $\phi_X(0) = E(\cos(0) + i\sin(0)) = 1$
2. $|\phi_X(t)| = \left|\int_\Omega e^{|tx|} dp\right| \leqslant \int_\Omega \left|e^{itx}\right| dp = \int_\Omega dp = 1 \quad \because |\cos(tx) + i\sin(tx)| = 1$
   $\therefore \quad |\phi_X(t)| \leqslant 1 \quad \forall t \in \mathcal{R}$
   Thus, $\phi_X(t)$ always exist. Unlike MGF, CF exists in every situation. Thus CF is stronger than MGR.
3. $\phi_X(-t) = \overline{\phi_X(t)}$ [conjugate].
   $$\because \phi_X(-t) = E(\cos(tx) + i\sin(-tx)) = E(\cos(tx) - i\sin(tx)) = \overline{\phi_X(t)}$$
4. If $\phi_X(t) = \overline{\phi_X(t)}$, $\phi_X(t)$ is real. Why?
   X is symmetrically distributed about zero.
   $\Leftrightarrow \phi_X(t)$ is real.
   $\because X \overset{d}{=} -X$ [X is identically distributed with $-X$]
   $$\begin{aligned}\phi_X(t) = E\left(e^{itx}\right) &= E\left(e^{it(-x)}\right) \\ &= E\left(e^{-itx}\right) \\ &= \phi_x(-t) = \phi_X(t)\end{aligned}$$
   i.e. $\phi_X(t)$ is real.

Let $X_1, X_2, \ldots X_n$ be independently distributed r.v.s with X_j has CF of $\phi_j(t)$. Define $S_n = \sum_{j=1}^n X_j$, CF of S_n is $\phi_{S_n}(t) = E\left(e^{itS_n}\right)$

$$\begin{aligned}= E\left(e^{it\sum_{j=1}^n X_j}\right) &= E\left(\prod_{j=1}^n e^{itX_j}\right) \\ &= \prod_{j=1}^n E\left(e^{itX_j}\right) \quad \text{(due to independence} \\ &= \prod_{j=1}^n \phi_j(t)\end{aligned}$$

Furthermore, if X_j's are iid with common CF $\phi(t)$, $\phi_{S_n}(t) = \{\phi(t)\}^n$
We can write the expansion,

$$\begin{aligned}\phi_X(t) &= E\left(e^{itX}\right) \\ &= E\left[\sum_{x=0}^{\infty} \frac{i^r t^r}{r!} x^r\right] = \sum_{r=0}^{\infty} \frac{i^r t^r}{n!} \mu'_r\end{aligned}$$

for all real t, provided $\mu'_r = E(X^r)$ exists for all $r \geqslant 1$. When $E\left(x^r\right)$ exists for $r = 1, 2, \ldots, n$, we have,

$$\phi_X^{(r)}(0) = i^r \mu'_r,$$
$$\phi_X(t) = \sum_{r=0}^{n} \frac{i^r t^r}{r!} \mu'_r + o\left(t^n\right)$$

where $\dfrac{o\left(t^n\right)}{t^n} \longrightarrow 0$ as $t \longrightarrow 0$.

5 Inversion Theorem

Levy inversion theorem

Let X be a RV with CF ϕ and DF F then,

$$\frac{1}{2\pi}\lim_{T\to\infty}\int_{-T}^{T}\left(\frac{e^{-ita}-e^{-itb}}{it}\right)\phi_X(t)dt$$
$$=P(a<X<b)+\frac{1}{2}[P(X=a)+P(X=b)], a<b$$

5.1 Furies inversion theorem

Let X be a rv with CF ϕ and df such that ϕ is absolutely integrable on $\mathcal{R}$ i.e. $\int_{-\infty}^{\infty}|\phi_x(t)|\,dt<\infty$ and $F'(x)=f(x)$ exists for all $x\in\mathcal{R}$. Then $f(x)=\frac{1}{2\pi}\int_{-\infty}^{\infty}e^{-itx}\phi_X(t)dt$.

Note: These inversion theorem represent that given a CF there exists a distribution function (or a density) corresponding to a RV. That is CF uniquely identify a distribution. Further in Levy inversion theorem if the PDF is continuous ie. if $P(X=a)=P(X=b)=0$,

$\frac{1}{2\pi}\lim_{T\to\infty}\int_{-T}^{+T}\left(\frac{e^{-ita}-e^{-itb}}{it}\right)\phi_X(t)dt=P(a<X<b)=f(b)-f(a)$.

Thus, we see that in both the theorem we can invert a CF to a distribution function. That is why the theorem is named so.

6 Continuity Theorem

Let $\{X_n\}$ be a sequence of rvs and X be another rv defined on the same prob space $(\Omega,\mathcal{F}_1,p)$. $\{\phi_n(t)\}$ be sequence of CF corresponding to $\{X_n\}$ and $\phi(t)$ is the CF of X. Then,

$$\phi_n(t)\to\phi(t) \text{ ar } n\to\infty.$$
$$\Rightarrow x_n\xrightarrow{\infty}x \text{ or } n\to\infty$$

$X_n\xrightarrow{\infty}X$ ie. the large sample dist of X_n is same as that of X.

Note: The above theorem is also hold for MGF provided all the moment generating function exists.

Ex. Find CF of Bin(n,p), Poi (λ), Geo(p).